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STRESS ANALYSIS MANUAL

by

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INTRODUCTION

This introduction serves a threefold purpose: (1) it summarizes the eleven chapters making up this manual; (2) it references several publications which contain related information; and (3) it lists the nomenclature most commonly used in the chapters.

CHAPTER SUMMARIES

Chapter 1 - Beams

Section 1.3 of this chapter presents the method of analysis for simple, shear web, and partial-tension-field beams in bending, as well as methods of determining the reactions on statically indeterminate beams. Section 1.4 treats beam columns, and Section 1.5 covers beams in torsion including helical springs.

Chapter 2 - Column Analysis

Section 2.2 of this chapter treats primary bending and torsional failure as well as crippling failure of columns of uniform cross section. Stepped and tapered columns are treated in Section 2.3, and the material on beam columns is in Chapter 1.

Chapter 3 - Bar Analysis

This chapter treats bars in tension with emphasis upon the effect of stress raisers.

Chapter 4 - Trusses

Section 4.3 of Chapter 4 gives methods of determining the stresses and deflections of statically determinate trusses, and Section 4.4 treats statically indeterminate trusses.

Chapter 5 - Frames and Rings

This chapter gives a general treatment of frames composed of straight elements of uniform cross section, in addition to particular solutions for various simple frames and circular rings under several types of loadings.

Chapter 6 - Plates

Methods for determining the critical buckling stress of both flat and curved plates with and without stiffeners and having various loadings are given. Charts and curves covering most common loadings and supports facilitate analysis.

β	empirical constant
θ	angle
Δ	increment or difference
δ	deflection
ϵ	strain
η	plasticity coefficient
θ	cylindrical coordinate
θ	angle or angular deflection
λ	empirical constant
λ	half wavelength of buckling
μ	Poisson's ratio
μ	torsional spring constant
μ_e	elastic Poisson's ratio
μ_p	plastic Poisson's ratio
ν	Poisson's ratio
ν_e	elastic Poisson's ratio
ν_p	plastic Poisson's ratio
ρ	radius of gyration
ρ	density
Σ	summation
ϕ	angle or angular deflection
ψ	angular deflection
Ω	empirical constant
ω	angular velocity

k	= diagonal tension factor
L	= length
L'	= effective length of beam
M	= applied bending moment
M _{cr}	= critical moment
M _{fp}	= fully plastic bending moment
M _{sb}	= secondary bending moment in flange
M _t	= bending moment due to transverse loads alone
M _y	= bending moment at the onset of yielding
m	= coefficient given by Figure 1-8
n	= number of active spring coils
n	= constant given in Section 1.3.1.6
o	= subscript, outside
P	= applied concentrated load
P	= axial load
P _u	= upright end load
p	= rivet spacing
p	= pressure
Q	= statical moment of cross section - $\int_{A_1} y dA$
q	= shear flow
q'	= beam shear at a distance of $2h/3$ from the beam end
q _r	= shear load of web to flange rivets (lb/in.)
q _t	= tension load on web to upright rivets (lb/in.)
q _{t'}	= increased tension load on web to upright rivets (lb/in.)
R	= reaction
r	= radius
r	= subscript, rivet
r _i	= inside radius
r _o	= outside radius
S	= tension force on edge of membrane (lb/in.)
s	= distance measured along curved path
s	= distance from centroidal axis to point of application of load
s	= subscript, shear
T	= torque
T _{max}	= maximum allowable torque
t	= thickness
t	= subscript, tension
t _e	= effective thickness
t _f	= flange thickness
t _s	= skin thickness
t _{st}	= thickness of closed stiffener
t _u	= upright thickness
t _w	= web thickness
U	= $L \sqrt{\frac{P}{EI}}$ for beam column

U	= developed length of elongated section
u	= subscript, ultimate
u	= subscript, upright
V	= shear force
W	= concentrated transverse load
w	= distributed transverse load
w	= subscript, web
x, y, z	= rectangular coordinates
y	= deflection of beam due to bending
y	= subscript, yield
α	= angle of diagonal tension
α	= coefficient given by Table 1-14
β	= coefficient given by Table 1-14
δ	= spring deflection
δ_s	= portion of spring deflection due to direct shear
η	= plasticity coefficient
ρ	= radius of gyration
θ	= slope of beam
Σ	= summation

1.3 Introduction to Beams in Bending

For the purposes of discussion, beams in bending are divided here into simple beams (Section 1.3.1) and shear web beams (Section 1.3.2). Shear web beams are further subdivided into shear resistant beams and partial tension field beams. If a beam is statically indeterminate, Section 1.3.4 must be consulted in order to determine the reaction forces and moments. Otherwise, the equations of statics may be used to determine the reactions.

1.3.1 Simple Beams in Bending

Simple beams in elastic and plastic bending are treated in Sections 1.3.1.1 and 1.3.1.3, respectively, while the possibility of lateral instability of deep beams in bending is treated in Section 1.3.1.5.

1.3.1.1 Simple Beams in Elastic Bending

This section treats simple beams in bending for which the maximum stress remains in the elastic range.

The maximum bending stress in such a beam is given by the formula

$$f_b = \frac{Mc}{I} \quad (1-1)$$

while the shear flow is given by

$$q = \frac{VQ}{I} \quad (1-2)$$

The membrane analogy also makes it apparent that stresses are very low at the ends of outstanding flanges or protruding corners and very high where the boundary is sharply concave. For example, Table 1-16 gives the stress concentration factor for the concave side of the shape in Figure 1-68. Multiplying the maximum stress obtained from the formula for thin rectangular sections in torsion by this factor gives the maximum stress on the concave side of a thin bent section.

TABLE 1-16

Stress Concentration Factor for Thin Sections in Torsion

r/t	1/8	1/4	1/2	1
Factor	2-1/2	2-1/4	2	1-3/4

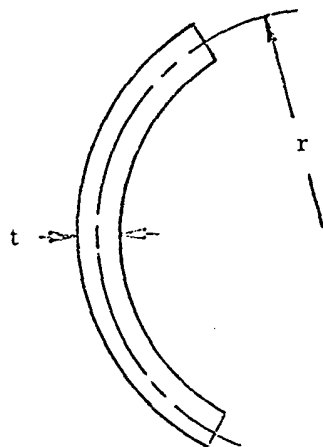


Figure 1-68. Thin Curved Section

1.5.3.2 Sand Heap Analogy for Beams in Plastic Torsion

The maximum ultimate torque that an open beam may withstand in torsion is given by

$$T = 2V F_{su} \quad (1-82)$$

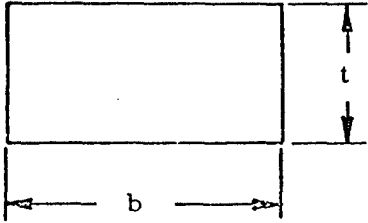
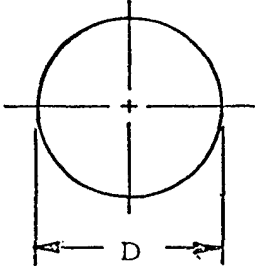
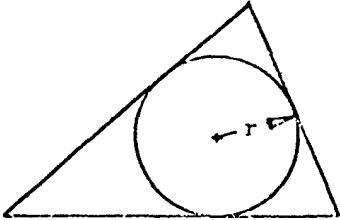
where V is the volume of a sand heap with a maximum slope of unity, piled on a plate having the same shape as the beam cross section. Table 1-17 gives the volume of sand heaps with various bases of various shapes.

1.5.4 Helical Springs

The primary stresses in the wire of a helical spring are due to torsion. Section 1.5.4.1 treats helical springs composed of round wire, and those composed of square wire are treated in Section 1.5.4.2.

TABLE 1-17

Sand Heap Volumes for Equation (1-82)

Type Section	Sand Heap Volume
Rectangle  A diagram of a rectangle. The horizontal width is labeled 'b' with a double-headed arrow below it. The vertical height is labeled 't' with a double-headed arrow to its right.	$V = \frac{t^3(3b-t)}{12} \quad b \geq t$
Circle  A diagram of a circle. A vertical line and a horizontal line intersect at the center, marked with a small cross. The horizontal line extends to the edges of the circle, and its length is labeled 'D' with a double-headed arrow below it.	$V = \frac{\pi D^3}{24}$
Triangle  A diagram of a triangle with an inscribed circle. The circle is tangent to all three sides of the triangle. A line segment from the center of the circle to one of the sides is labeled 'r' with a double-headed arrow. A. = area of triangle r = radius of inscribed circle	$V = \frac{Ar}{3}$

1.5.4.1 Helical Springs of Round Wire

Figure 1-69 shows a helical spring made of round wire under an axial load, P . If the spring radius (r) is much greater than the wire diameter (D), the wire may be treated as a straight round beam under a torsional load, Pr , as indicated in Figure 1-69. Superposing the stress due to torsion of the wire on the uniform shear stress due to direct shear ($4P/\pi D^2$), the following equation for the maximum shear stress in the spring may be obtained:

$$f_{\max} = \frac{16 Pr}{\pi D^3} \left(1 + \frac{D}{4r} \right) \quad (1-83)$$

In the cases of heavy coil springs composed of wire with a relatively large diameter, D , in comparison to r , the initial curvature of the spring must be accounted for. This is done in the following equation:

$$f_{\max} = \frac{16 Pr}{\pi D^3} \left(\frac{4m-1}{4m-4} + \frac{0.615}{m} \right) \quad (1-84)$$

where

$$m = \frac{2r}{D} \quad (1-85)$$

This equation reduces to Equation (1-83) as r/D becomes large.

The total deflection (δ) of a round spring of n free coils is given by

$$\delta = \frac{64 Pr^3 n}{GD^4} \quad (1-86)$$

This equation neglects the deflection due to direct shear which is given by

$$\delta_s = \frac{8 PRn}{Gd^2} \quad (1-87)$$

This portion of the deformation, however, is generally negligible compared to the value of δ given by Equation (1-86) and is thus generally ignored.

All of the equations in this section apply to both compression and tension springs, and in both cases the maximum shear stress occurs at the inside of the wire.

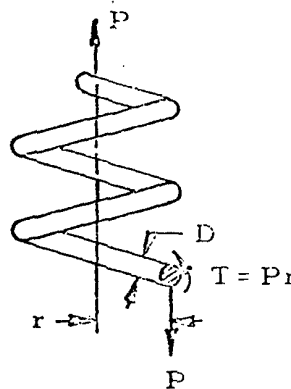


Figure 1-69. Helical Spring of Round Wire

1.5.4.2 Helical Springs of Square Wire

Figure 1-70 shows a helical spring made of square wire under an axial load, P . The maximum shear stress in the square wire is given by

$$f_{s_{max}} = \frac{4.80 Pr}{b^3} \left(\frac{4m-1}{4m-4} + \frac{0.615}{m} \right) \quad (1-88)$$

where

$$m = \frac{2r}{b} \quad (1-89)$$

The total deflection of such a spring is given by

$$\delta = \frac{44.5 Pr^3 h}{Gb^4} \quad (1-90)$$

where n is the number of active or free coils in the spring. This equation neglects the deflection due to direct shear as did Equation (1-86). However, the deflection due to direct shear is normally negligible compared to that given by Equation (1-90).

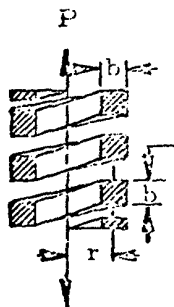


Figure 1-70. Helical Spring of Square Wire